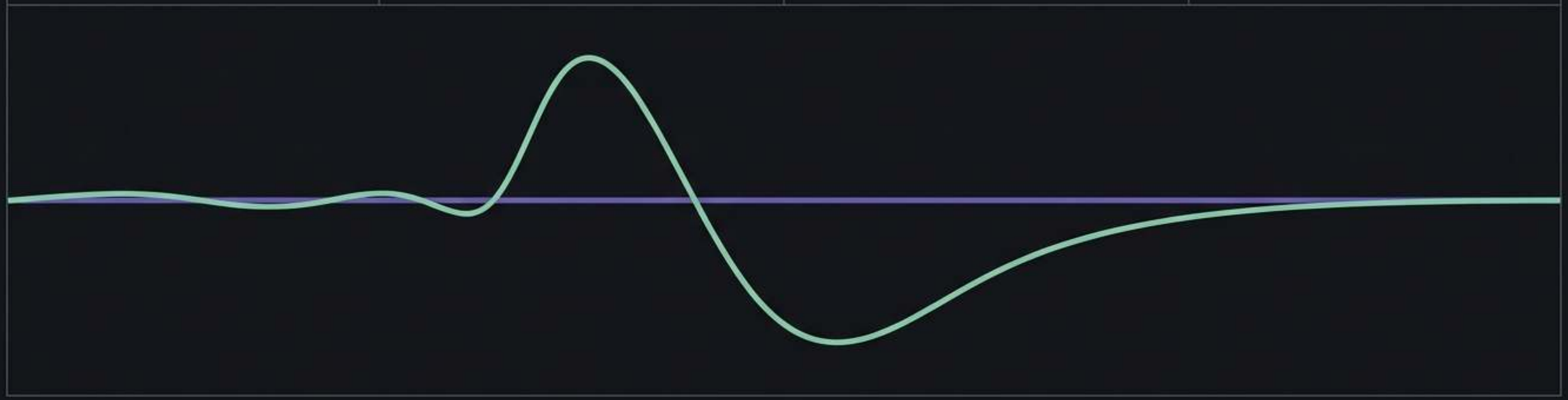
The Coherence Observatory

Examining the unseen relationships within systems
via The Coherence Engine™ Research Portfolio.



The Core Premise

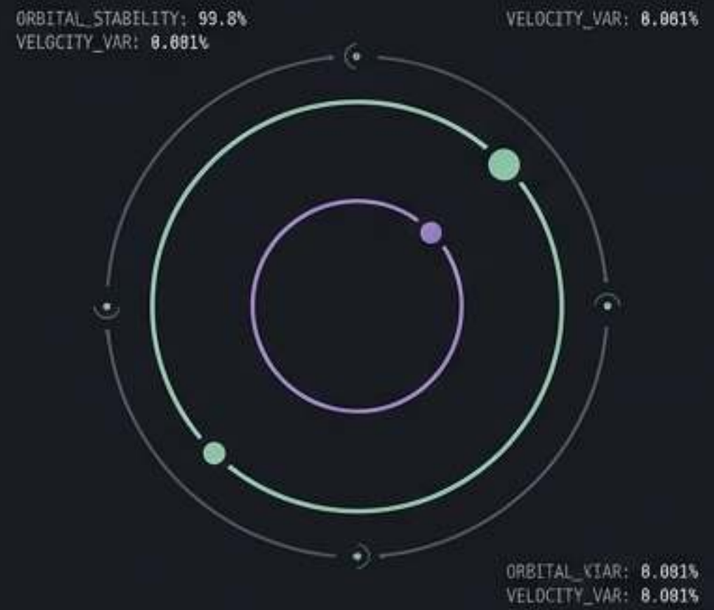
See what individual signals cannot reveal.

The Focus

Identifying drift, instability, and recovery across industrial, biological, and computational networks.

Gemini Notebook

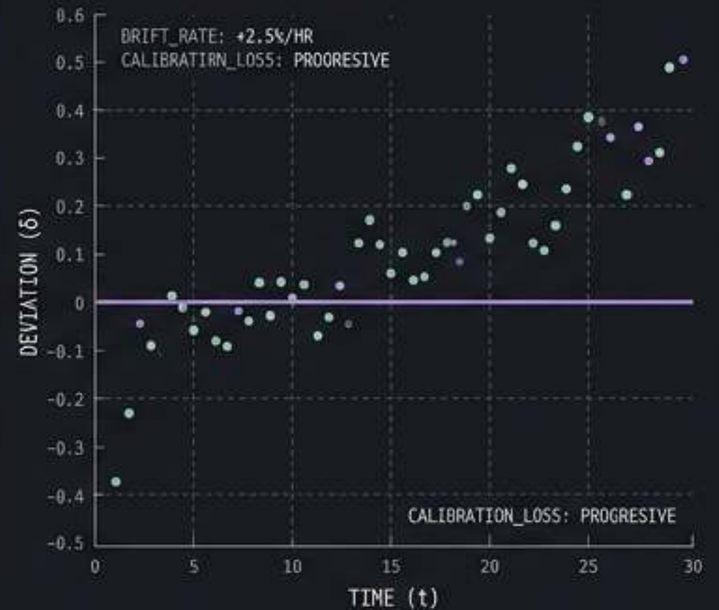
Managing systemic change requires analyzing relationships, not just thresholds.



01 Prioritize Reviews

Identify which systemic changes represent true anomalies deserving immediate attention versus acceptable noise.

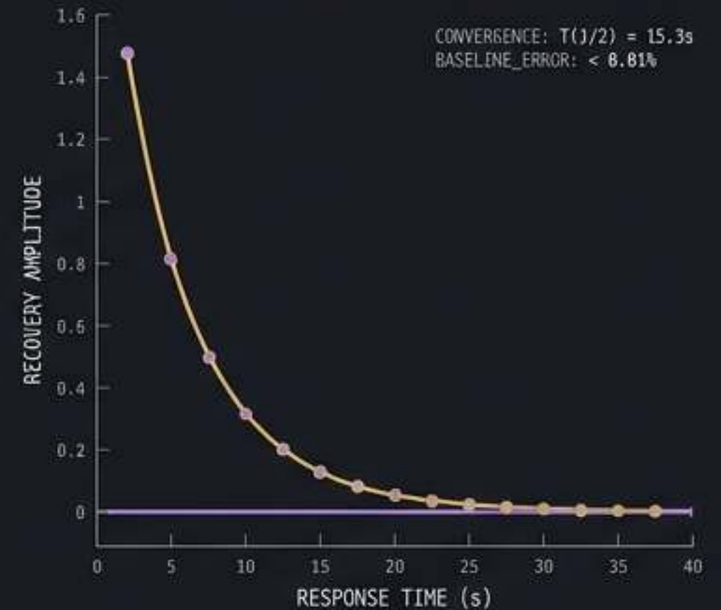
PRIORITY_LEVEL: CRITICAL_THRESHOLD > 5σ



02 Examine Intervention Burden

Balance the frequency of corrective actions against computational or physical resource utilization.

RESOURCE_COST: COMPUTATION = 4.2e6 FLOPs



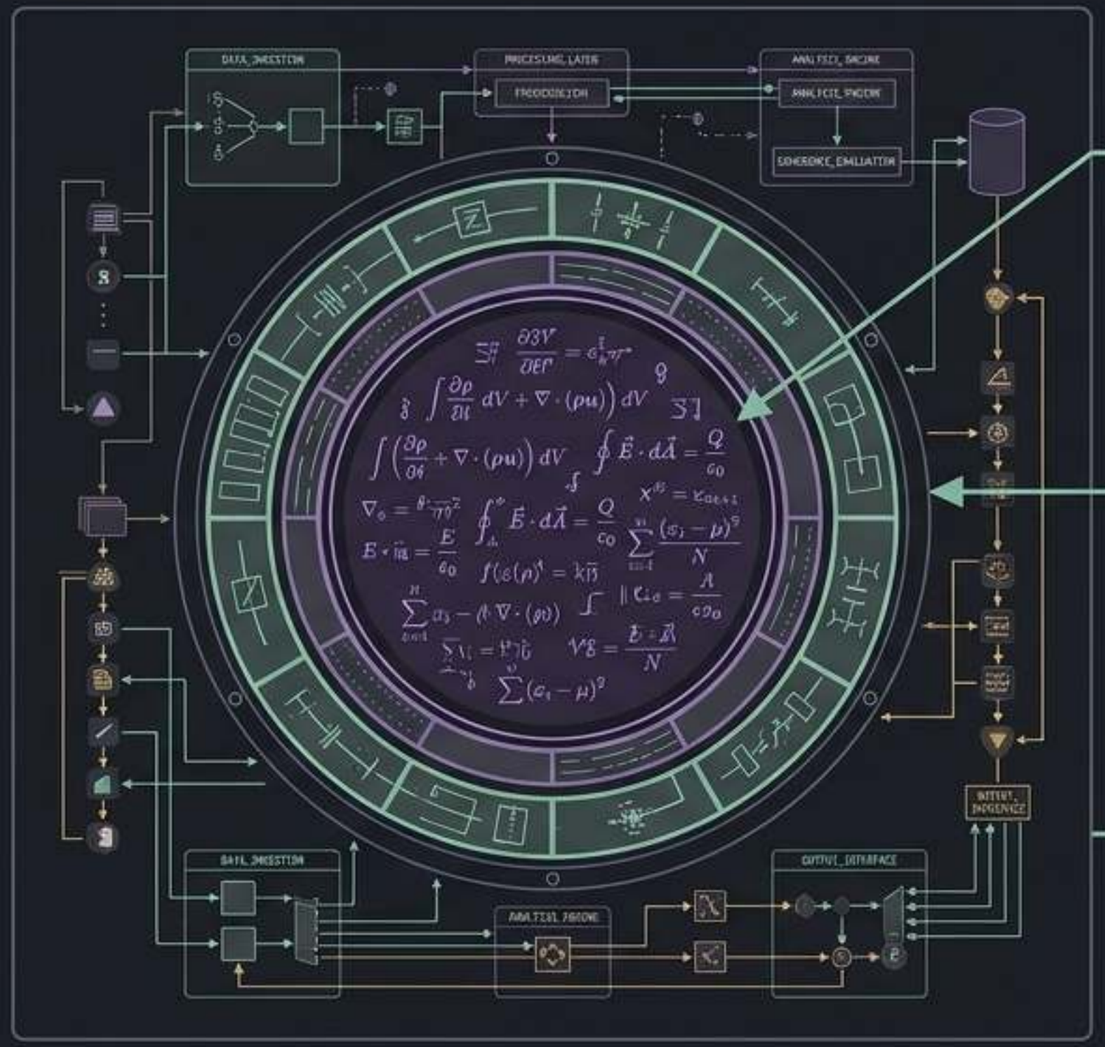
03 Track Recovery

Monitor a system's precise trajectory and time-to-baseline as it returns toward its reference state.

TRAJECTORY_STATUS: STABLE_CONVERGENCE

Gemini Notebook

One mathematical formulation drives the implemented computational framework.



Center - Δ.72 Unified Field Equation

The authored, canonical mathematical formulation (documented October 2025).

Middle - Δ.72 Coherence Framework

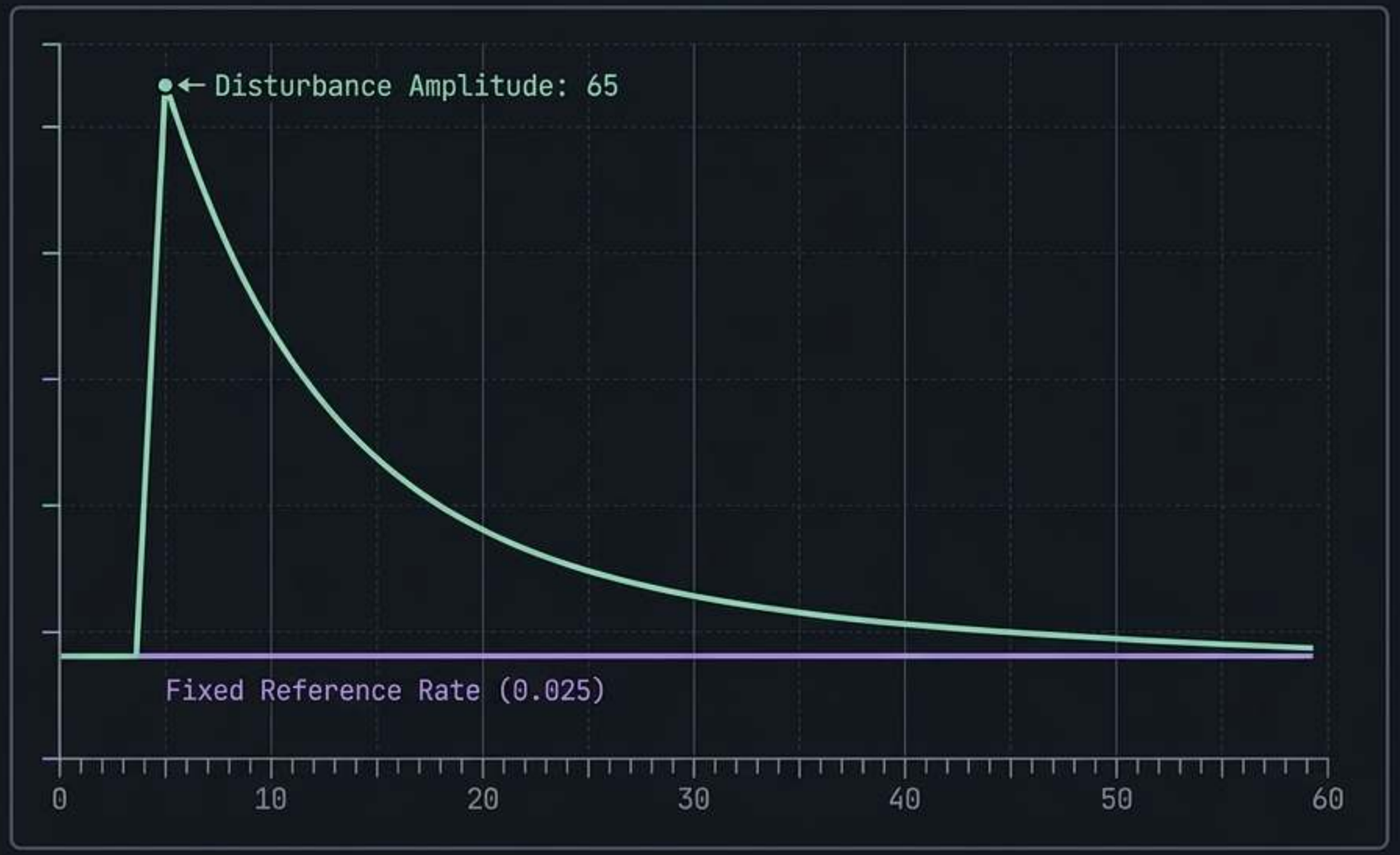
The structural definitions and operators for relational dynamics, degradation, and system recovery.

Outer Ring - The Coherence Engine™

The benchmark-tested, executable computational layer that evaluates these changing relationships in active environments.

Gemini Notebook

Following a system's return toward its reference reveals hidden resilience.



Selected Recovery Rate:
0.087 per step

Selected Half-Life:
8.0 steps
(vs reference 27.7 steps)

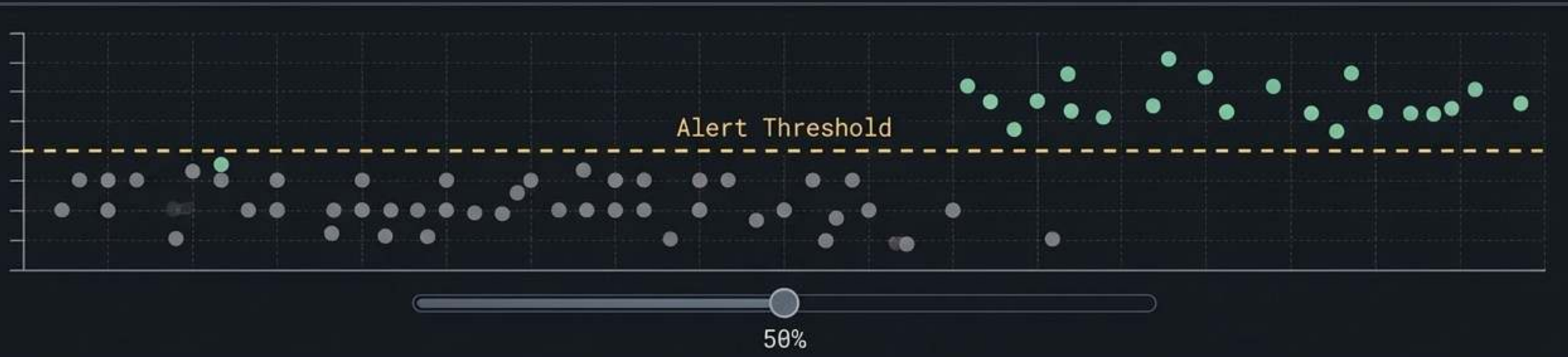
Remaining Deviation at step 59:
0.004

Operational Insight: Similar initial disturbances yield drastically different recovery trajectories. Tracking this decay frames critical questions about observation timing and intervention.

(Disclaimer: Synthetic illustration of exponential decay, not real-world claim).

Gemini Notebook

Balancing the intervention burden requires strict threshold calibration.



The Scenario

60 synthetic readings, 12 labeled positives, 48 negatives. Detector separation: 45%.

At 50% Threshold

12	Detected events	0	Missed events	12	False alerts

The Tradeoff

Lowering the threshold catches more events but increases review workload.

Raising it saves time but accepts the risk of missed failures.

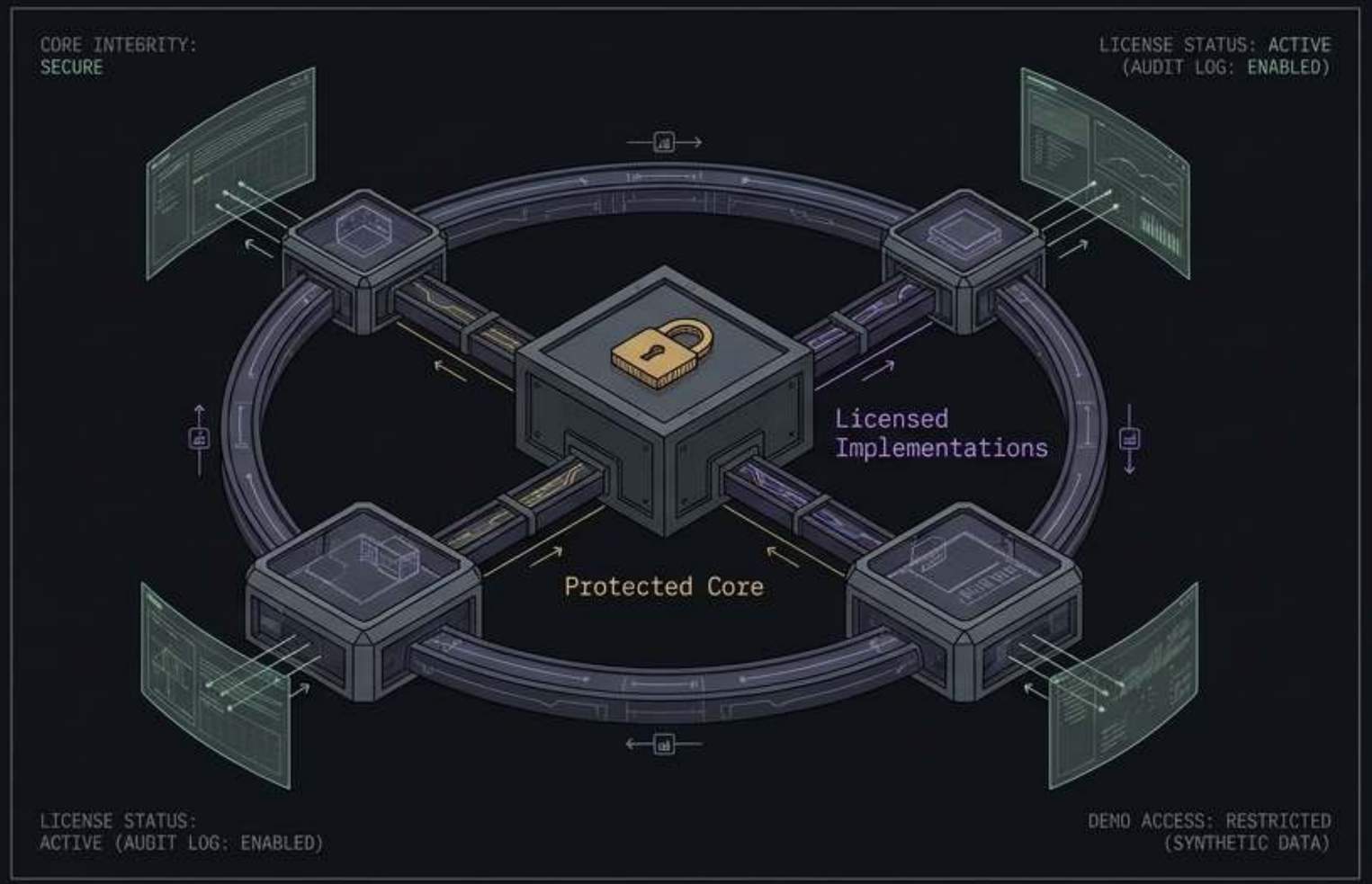
Gemini Notebook

Empirical benchmarks validate utility across distinct operational domains.

DOMAIN	COMPARATOR	OUTCOME	RELEVANCE
Scania APS (Logistics)	Always-on Extra Trees	85.7% fewer heavy-model calls (13,708 avoided); false positives dropped 841 → 740; 740; missed failures unchanged (6).	Reduced evaluation workload while preserving failure detection.
HBI v0.4.1 (Energy Control)	Locked baseline control (72 cases)	20.2% less modeled energy, 88% fewer commands, 0 violations.	Demonstrated tradeoff between control activity and resource use.
Hydraulic Fault Diagnostics (Industrial)	EWMA-only monitoring (UCI rig)	AUROC 0.757 vs 0.640. Flagged 16/45 positive blocks vs EWMA's 8/45.	Added diagnostic value alongside existing industrial monitoring.

Gemini Notebook

Domain-specific digital twins protect the computational core.



Protected Core

Operator logic, weights, calibration rules, and the proprietary Engine remain entirely protected and offline.

Licensed Implementation

Field-specific twins (e.g., NASA C-MAPSS, Industrial Compressors) licensed for evaluation under strict scope, audit rights, and claim controls.

Public Showroom

Synthetic demos, concept architectures, and benchmark summaries engineered for diligence and partnership discussions.

Gemini Notebook

The implementation spans four primary application environments.



Industrial + Infrastructure

Predictive maintenance, grids, pipelines, automated rail, semiconductor fabs, and data-center runtime governance.



Human + Biological

Public health surveillance, ECG arrhythmia tracking, Coherence Sound™, and Food Coherence System (FCS) measurement pathways.



Enterprise + Decision

RecoverMargins, legal/operations drift, business margin recovery, and AI learning progression governance.

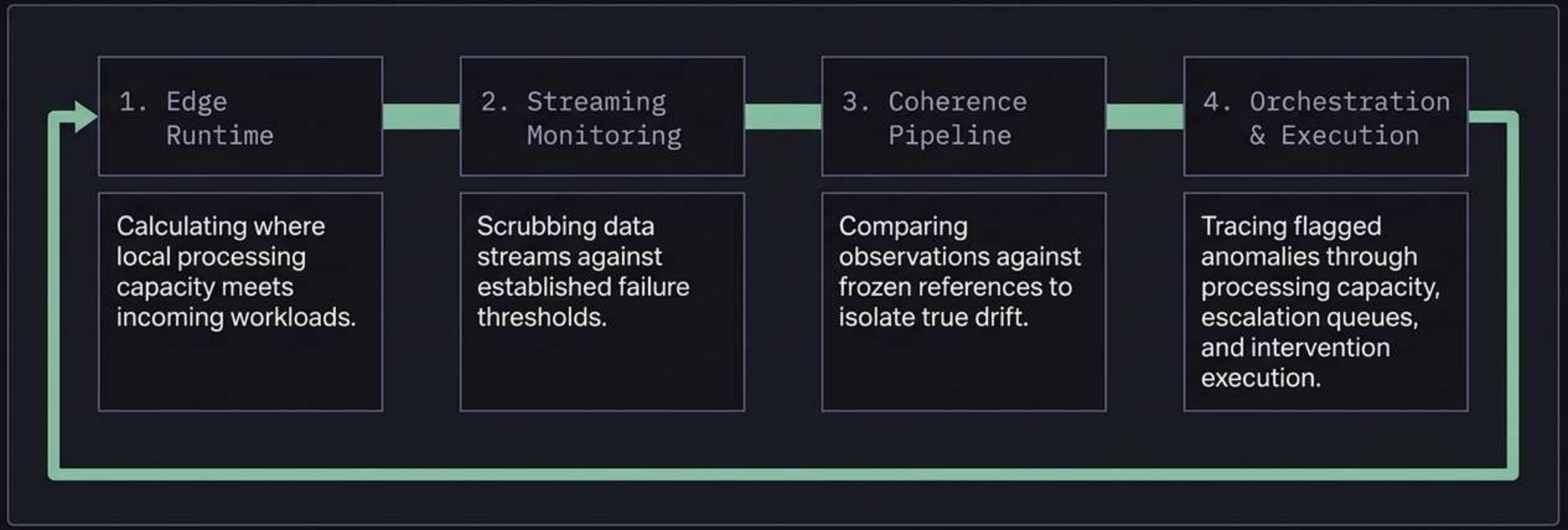


Regenerative + Frontier

Biohybrid substrates, regenerative closed-loop buildings, quantum calibration drift, and physical-engine concepts.

Gemini Notebook

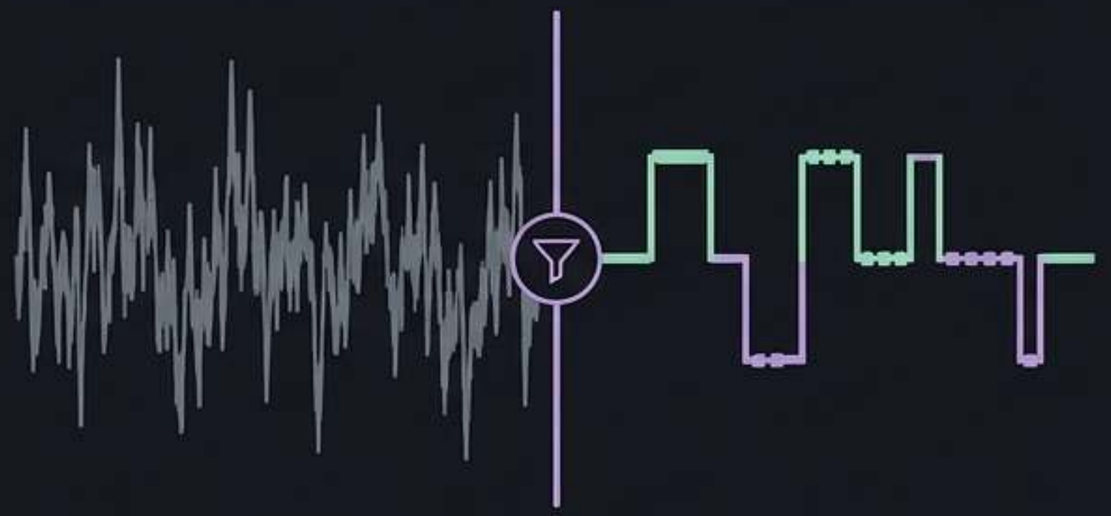
Signals move from local edge execution to systemic orchestration.



Gemini Notebook

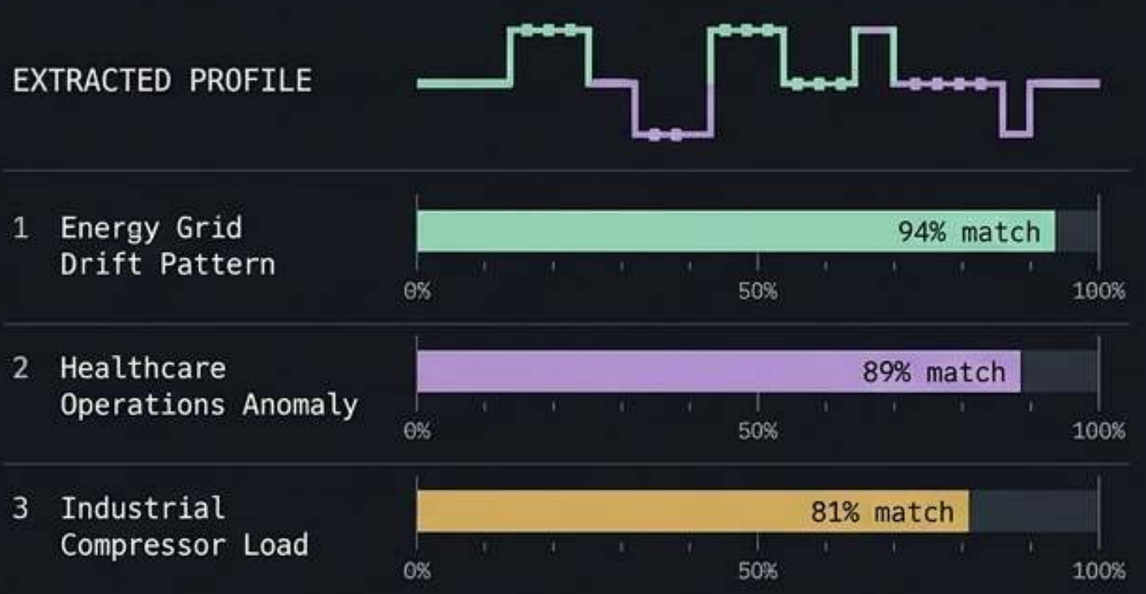
Pattern extraction enables cross-system adaptive matching

Pattern Engine



RAW SESSION DATA > FILTERING > STRUCTURAL PROFILE

Cross-System Similarity



Pattern Engine & Signature Library

Extracts structural profiles from session data and compares feature vectors against illustrative instability templates.

Cross-System Learning

Identifies operational similarities between highly disparate contexts.

Governance & Audit

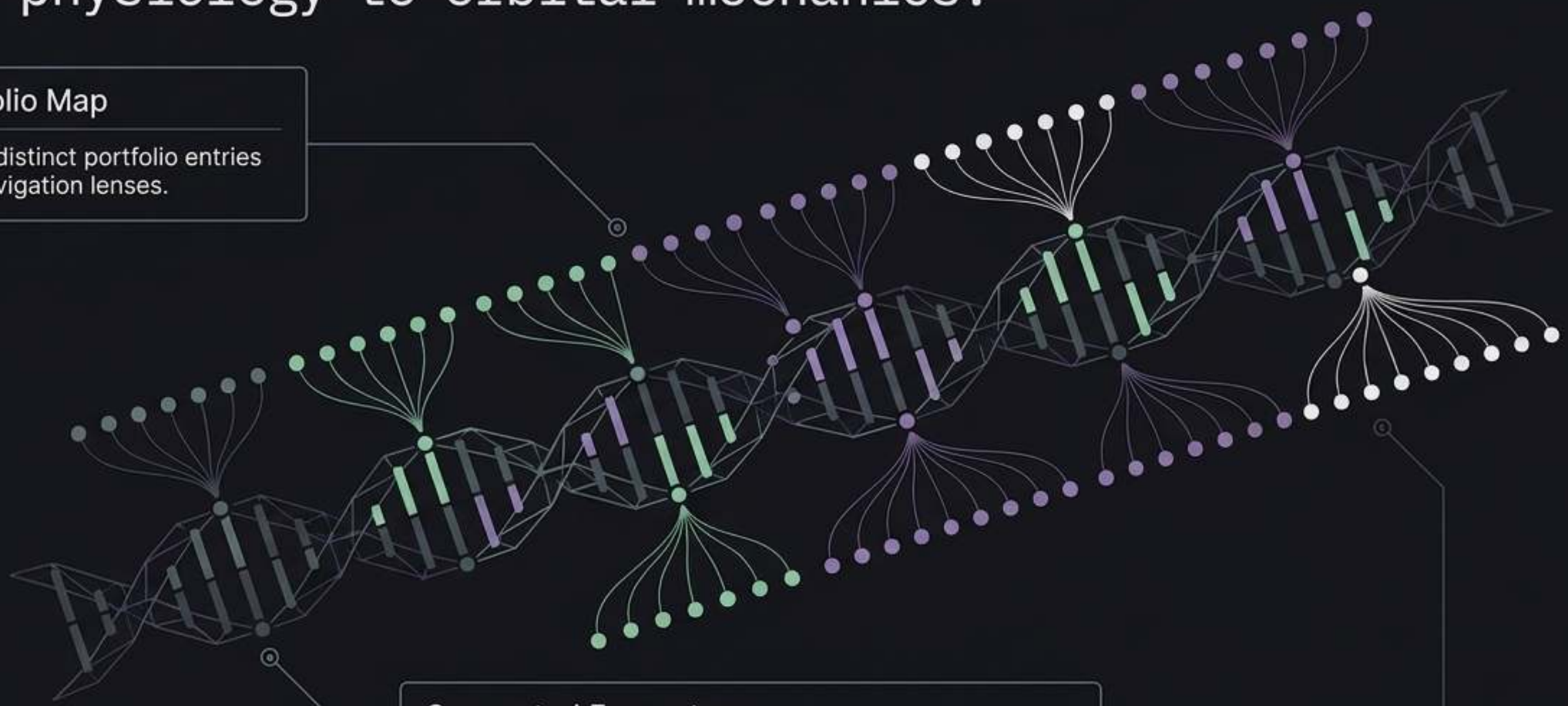
Strict, opt-in sharing boundaries ensure continuous learning without compromising tenant data sovereignty.

Gemini Notebook

A single computational grammar connects human physiology to orbital mechanics.

The Portfolio Map

Tracking 64 distinct portfolio entries across 12 navigation lenses.



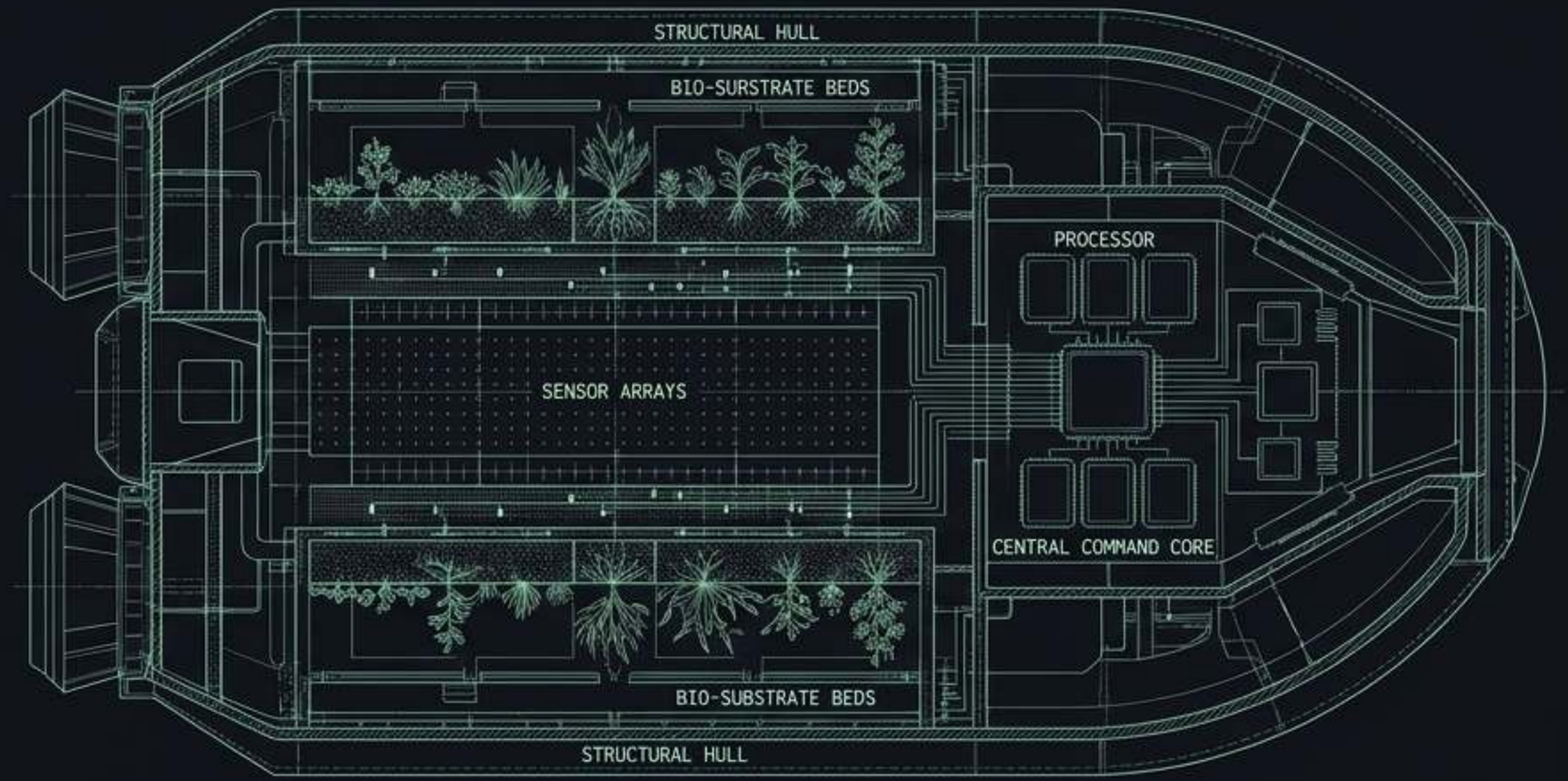
Connected Ecosystem

Explaining how the identical mathematical formulation of the Δ.72 Coherence Framework scales from tracking computational energy efficiency to evaluating biological signals and designing closed-loop habitats.

Note: Strictly delineates benchmarked computational work from symbolic/proposed inquiry.

Gemini Notebook

Extreme frontiers demand coherence-governed closed-loop infrastructure.



Bio-Integrated Autonomous Infrastructure Platform

Integrating biological substrates, advanced sensors, computing, and coherence-governed supervision into a single living system.

Coherence-Governed Spacecraft

Proposed architecture combining biological subsystems and closed-loop resource monitoring, managed entirely by The Coherence Engine™ for crewed transit missions (e.g., Mars habitats).

Physical Engine Research

Exploration of resonance chambers, momentum-coupling hypotheses, and Δ.72 Coherence-Controlled physical actuators.

Gemini Notebook

Allison Hensgen architectures a unified approach to physical and computational systems.

THE PROFILE

Allison Hensgen

Interdisciplinary systems architect and founder of the Δ.72 Coherence Framework.

ORCID: 0009-0008-7247-0307
Scholar: Coherence Systems Archive

THE ARCHIVE



Canonical Manuscripts

Open-repository foundation papers including “The Coherence Field: A Generative Model for Coherence-Driven Physical Law” and “Time as Vibration”.

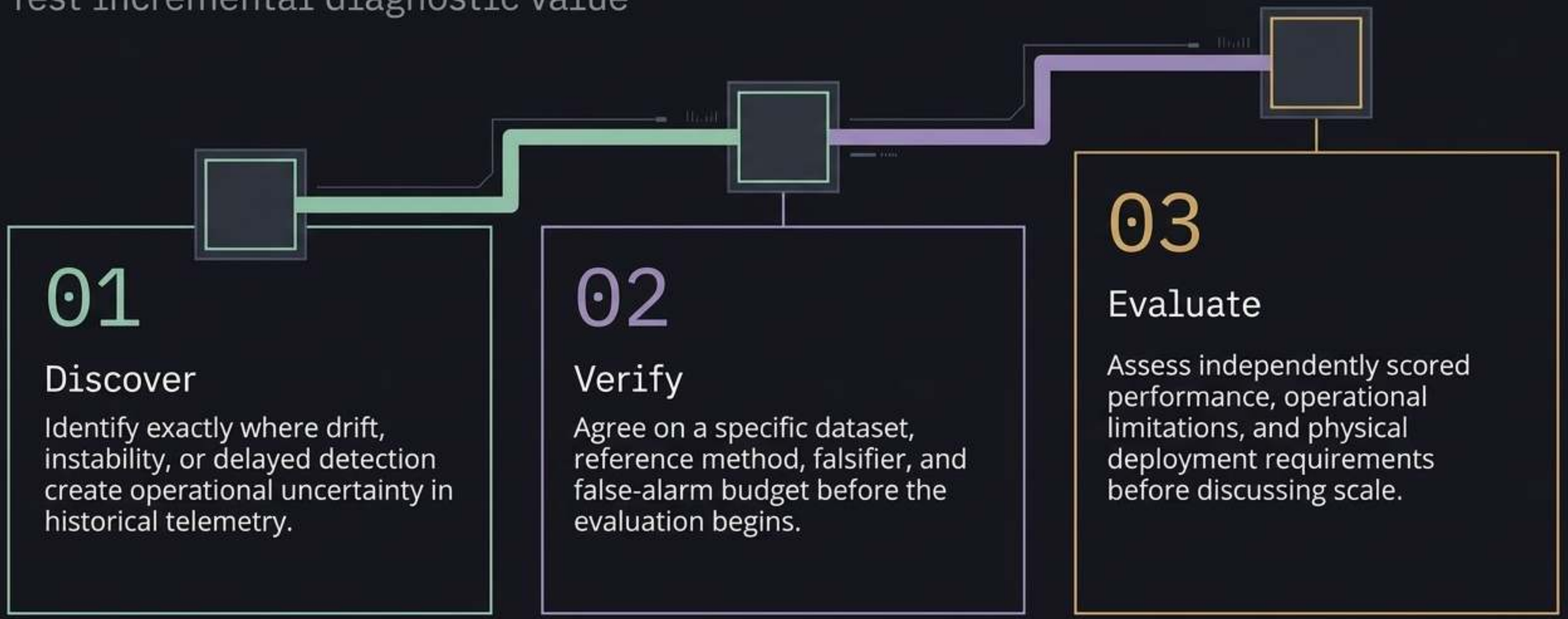
Research Communication

Establishing broader understanding via the Coherence Physics communication channels (delineating conceptual explanations from experimental results).

Gemini Notebook

Establishing incremental diagnostic value requires strict, scoped evaluation.

Test incremental diagnostic value



01

Discover

Identify exactly where drift, instability, or delayed detection create operational uncertainty in historical telemetry.

02

Verify

Agree on a specific dataset, reference method, falsifier, and false-alarm budget before the evaluation begins.

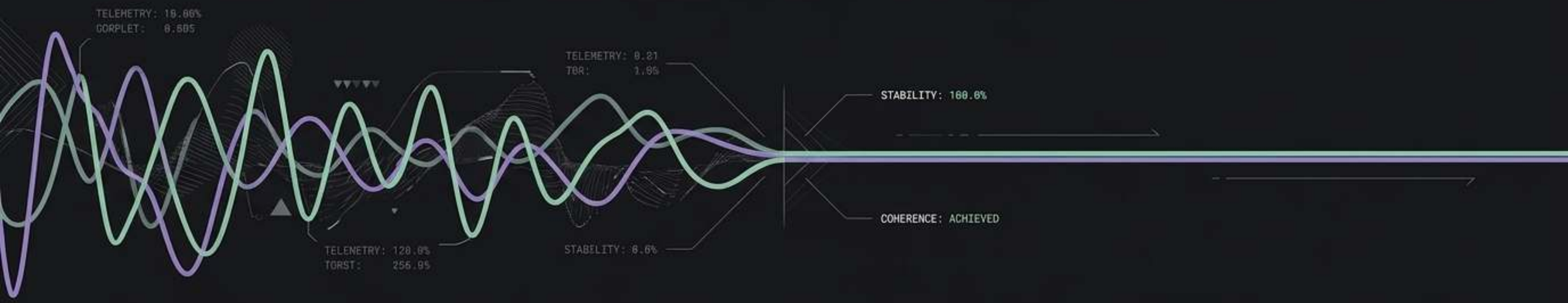
03

Evaluate

Assess independently scored performance, operational limitations, and physical deployment requirements before discussing scale.

Gemini Notebook

Choose the operational consequence that matters most.



The Core Choice

Is your priority reducing review workload, preventing missed events, lowering intervention burden, or accelerating recovery time?

Next Steps

- Explore the Evidence Repository & Interactive Replays
- Define a Scoped Industrial Pilot
- Request Licensed Digital Twin Access

Contact: CoherenceDashboard@gmail.com

Gemini Notebook